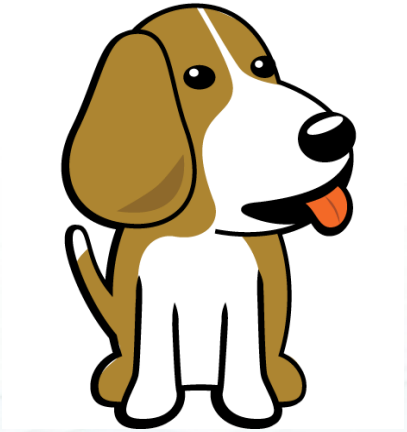
BeagleDust

BeagleDust

Concept and business case
Premier Farnell and BeagleBoard.org
December 2018

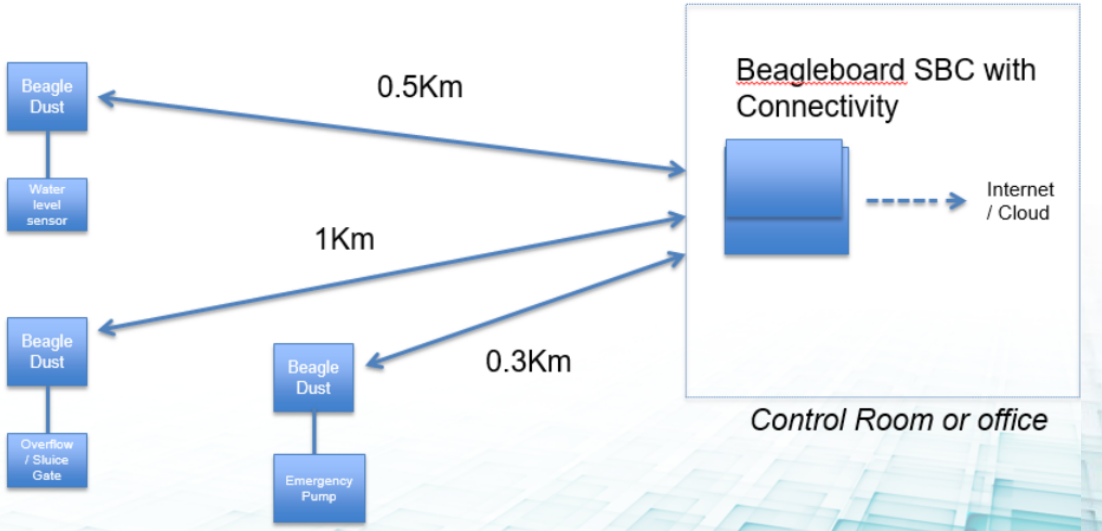


The Concept

element14

- **Digital Dust ->** A low cost sub $10, low-profile edge node that is easily deployed almost anywhere
  - Sub 1GHz network based on TI-15.4 stack
- User experience is simple and seamless with out of box setup taking minutes

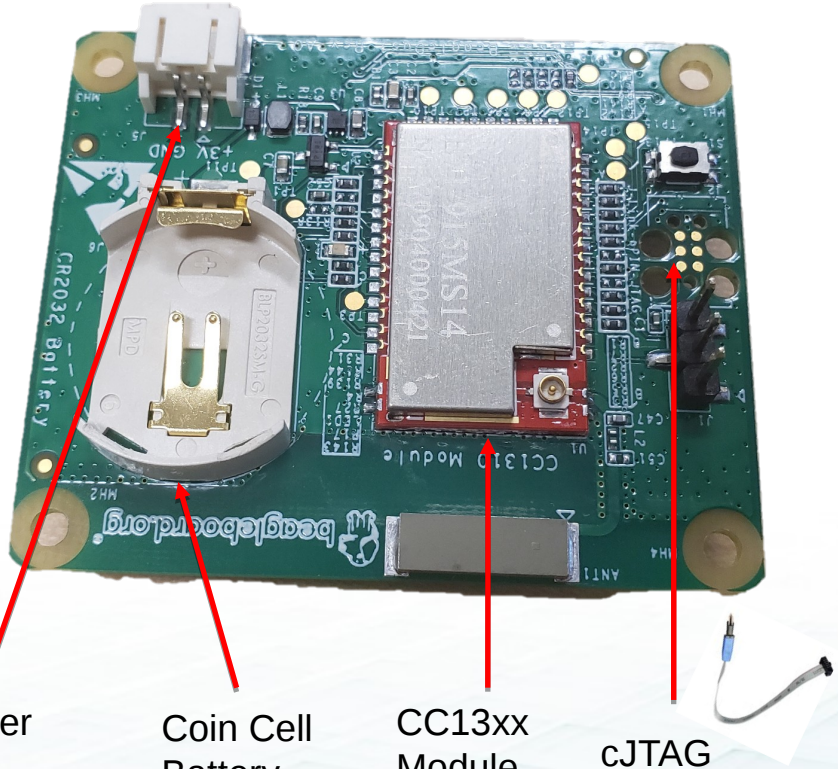
- Adaptable by Makers and Professionals alike for it's plug and play usability
  - Professionals would use for proof of concept, assume higher volumes would result in design / manufacturing customization opportunities and the wider portfolio of IoT services



BeagleDust Feature Set

element14

FRONT



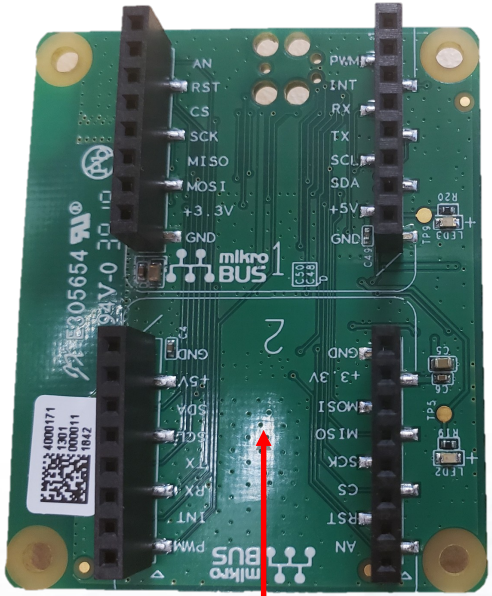
Power

Coin Cell Battery

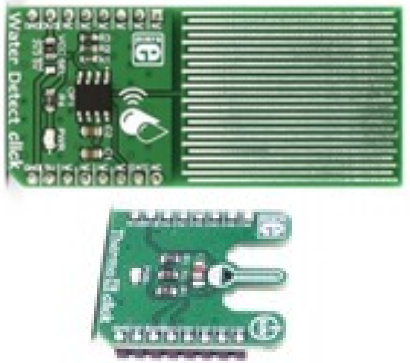
CC13xx Module

cJTAG

BACK



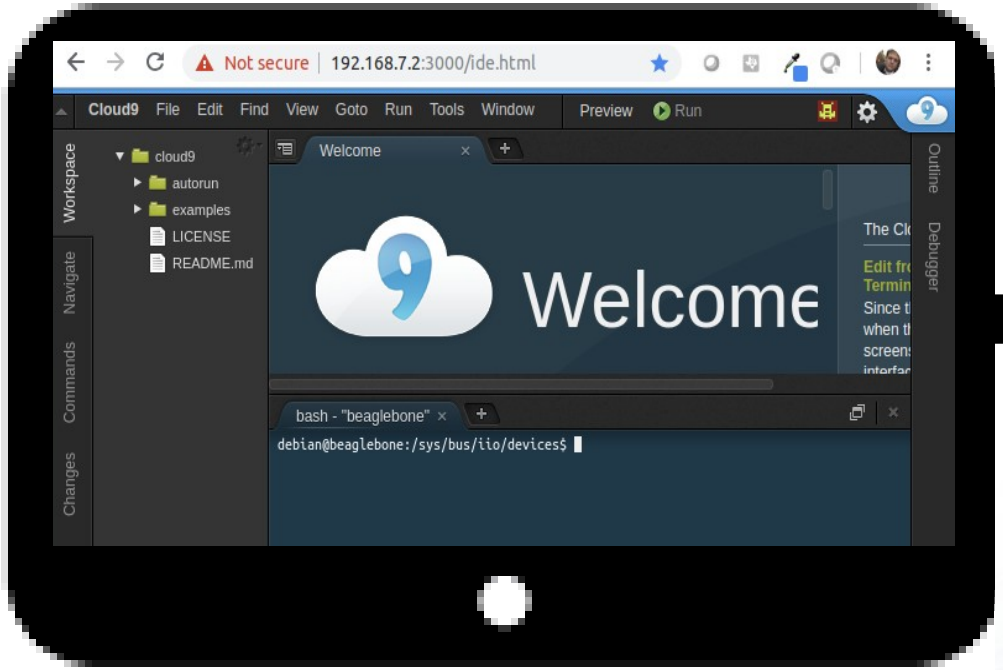
2x mikroBUS Headers



Click Boards

The User Experience

Step 1 – Gateway login



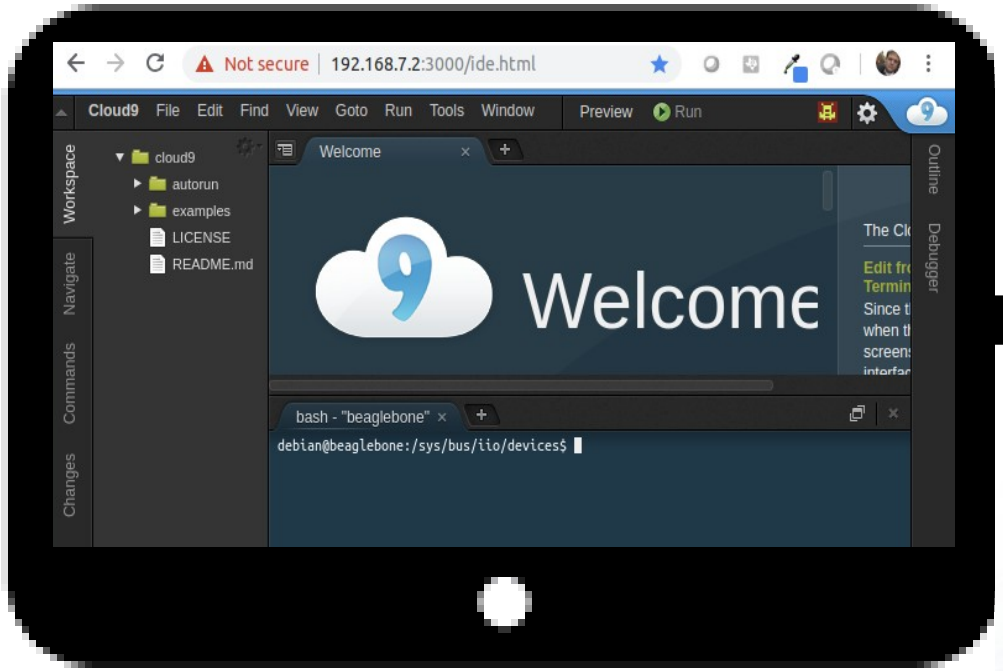
IDE

Gateway

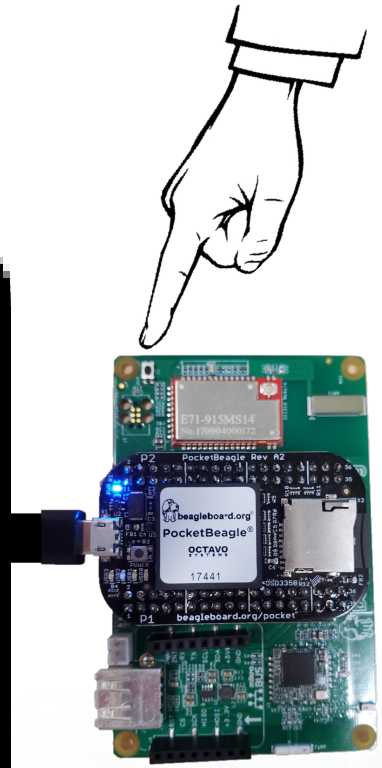
SensorTag

The User Experience

Step 2 – Connect with button push



IDE

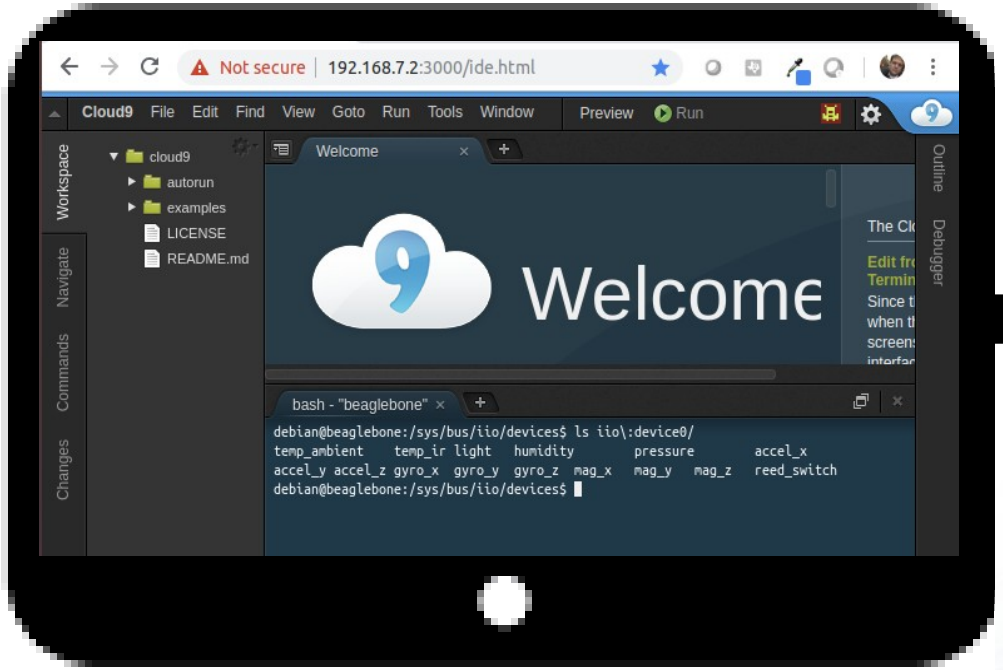


Gateway

SensorTag

The User Experience

Step 3 – Live edge data automatically appears



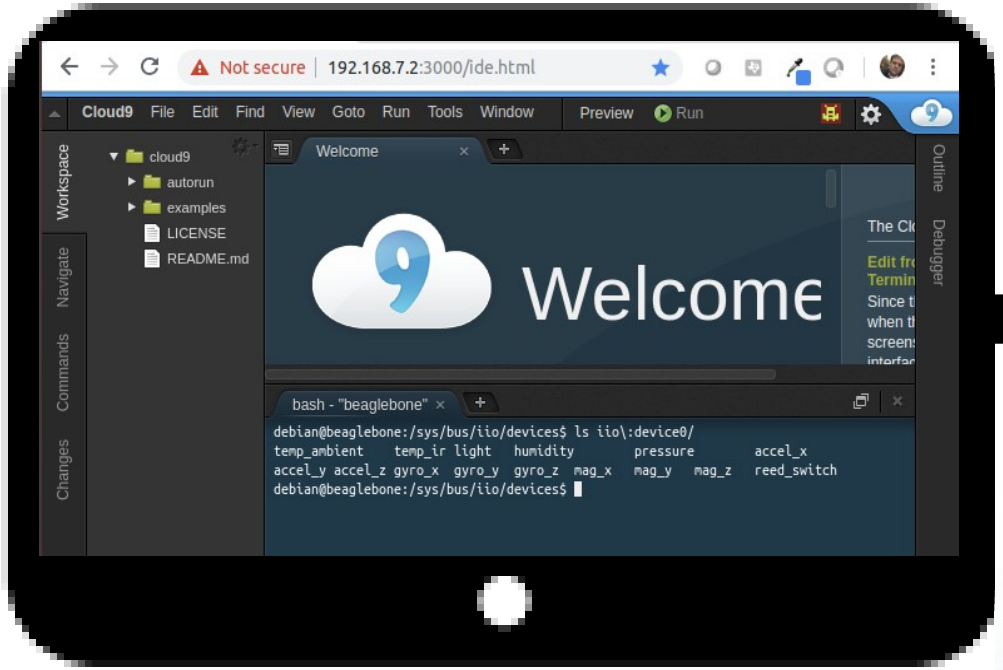
IDE

Gateway

SensorTag

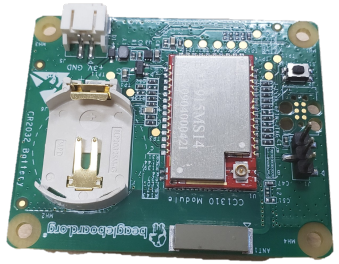
The User Experience

Same firmware, infinite sensors



IDE

Gateway



BeagleDust

SensorTag

User Experience Value Proposition

1. Automatic provisioning for I2C, SPI, GPIO, UART, ADC, and PWM
2. No cut-and-paste
    1. No device specific code lives my program
    2. Driver maintenance is owned by Linux kernel
3. No “firmware” development
    1. Everything a developer needs to do sits above the OS in any language
4. All sensor data is readily accessible in read and write files
5. Every programming language supported
    1. No language specific libraries
6. Out of the box support for ecosystem of 500+ mikroElektronika click boards

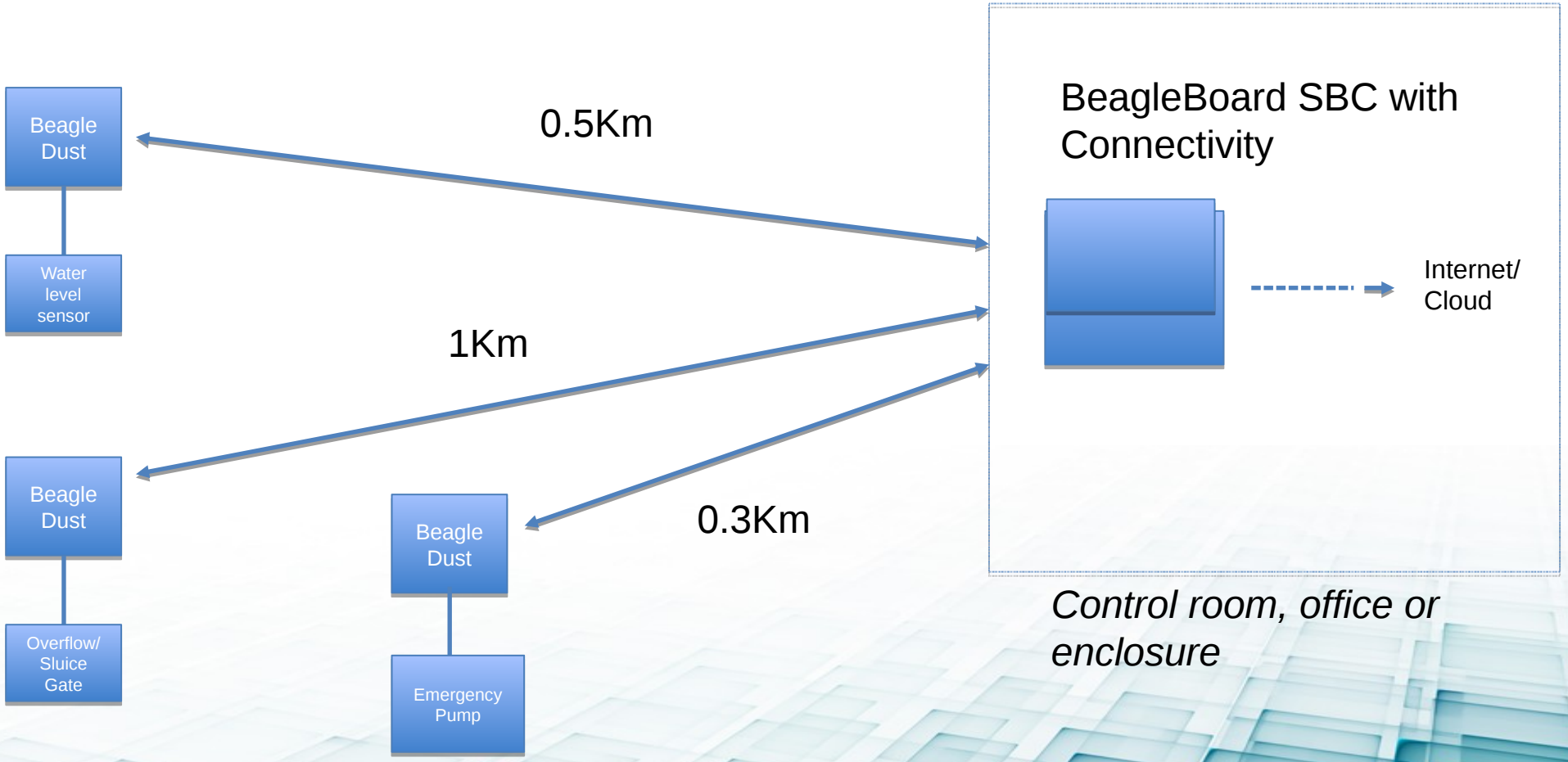
The Ask

1. Marketing Support
    1. How can TI drive market awareness – what does TI need from BB.org to do this?
2. Software and Hardware Development
    1. Design support and consultancy for discrete implementation of CC13xx
        1. Required to achieve pricing targets
    2. Split cost of 6 months firmware development, ~$150k
3. Pricing
    1. Target BOM cost is $3 to achieve a MSRP of <$10

Appendix

Example: flood mitigation system

element14

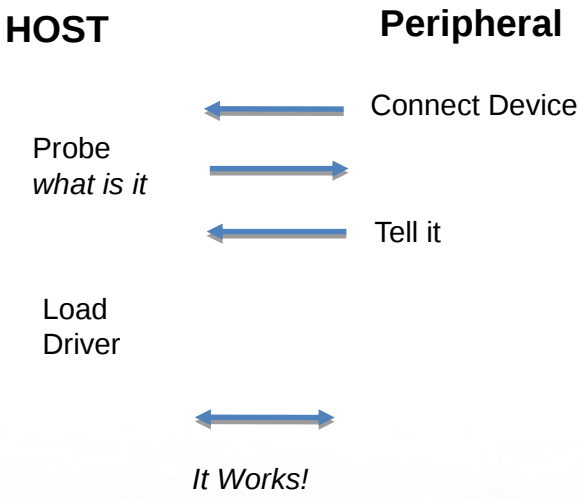


Software Proposition

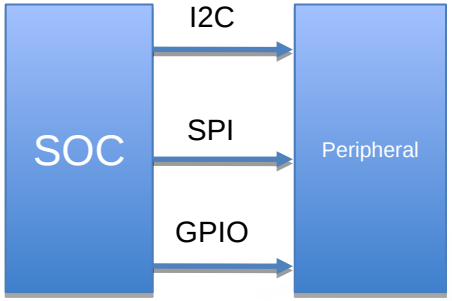
element14

- Greybus: automatic provisioning for I2C, SPI, and GPIO

USB in Linux Today

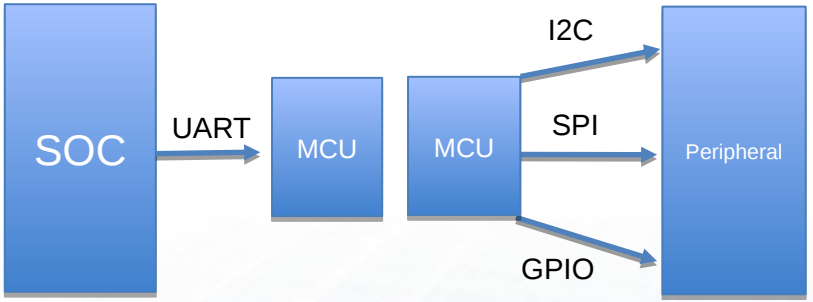


Before Greybus



Specific device tree required

With Greybus



The Concept

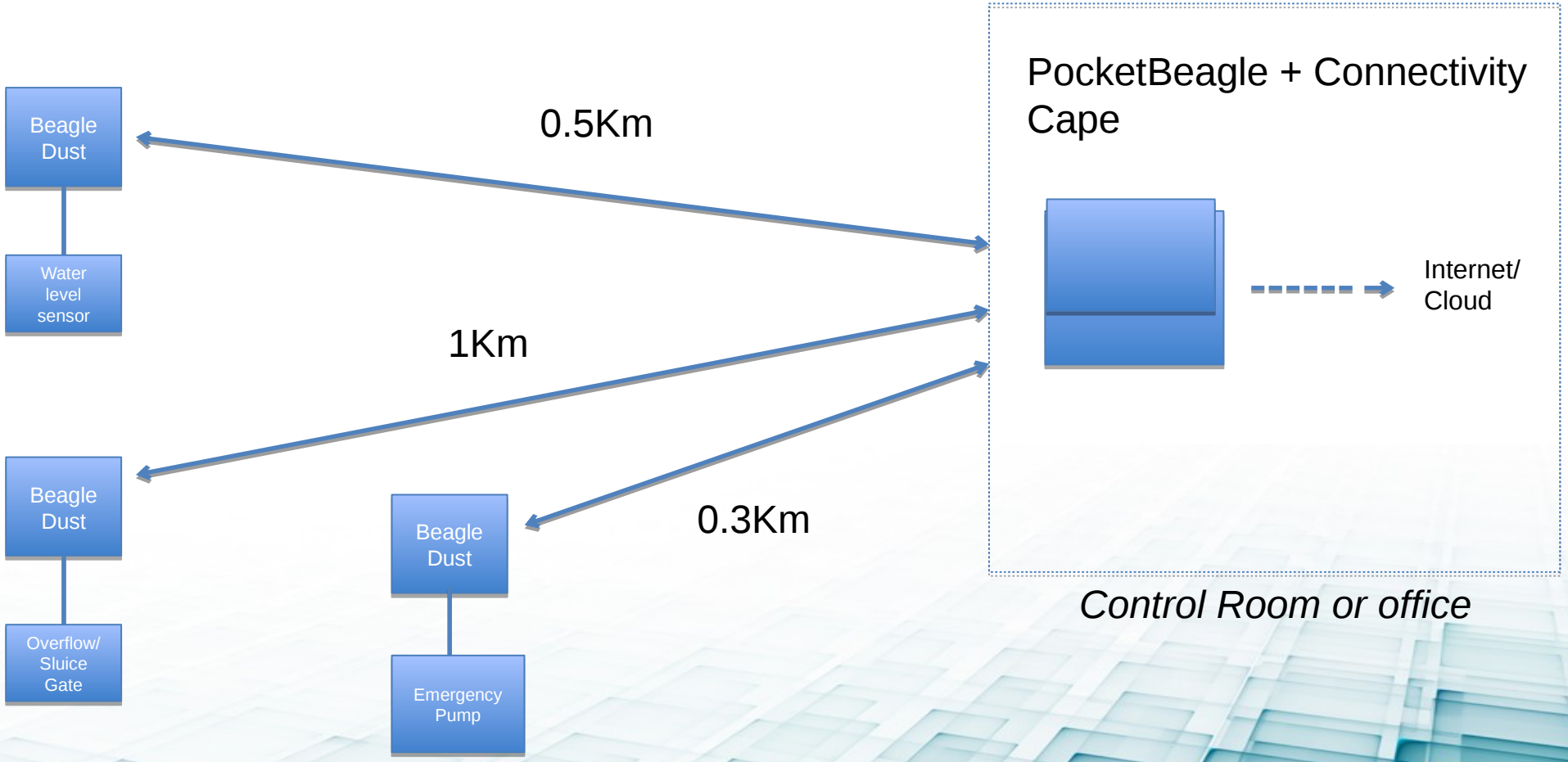
element14

- Low cost (< $10), low-profile edge node that is easily deployed almost anywhere
- Adaptable by Makers and Professionals alike for it’s plug and play usability
  - Professionals would use for proof of concept, assume higher volumes would result in design / manufacturing customization opportunities and the wider portfolio of IoT services
- Features Long Range Radio comms (Sub-GHz TI 15.4 compare with LoRA) module with on-board MCU for control and uFL connector
- Low power with ample GPIO for easily sensor attachments
- Compatible with MikroBus interface, allowing for deployment of MikroE click sensor & actuator boards
- Works (& communicates) out-of-the-box with PocketBeagle or BBB, enabled with their respective Long Range Comms cape
- Debug / Programming via header / JTAG connector / USB option(s)
- Battery-holder for Li-Ion coin-cell
- Power-management with appropriate voltage translations

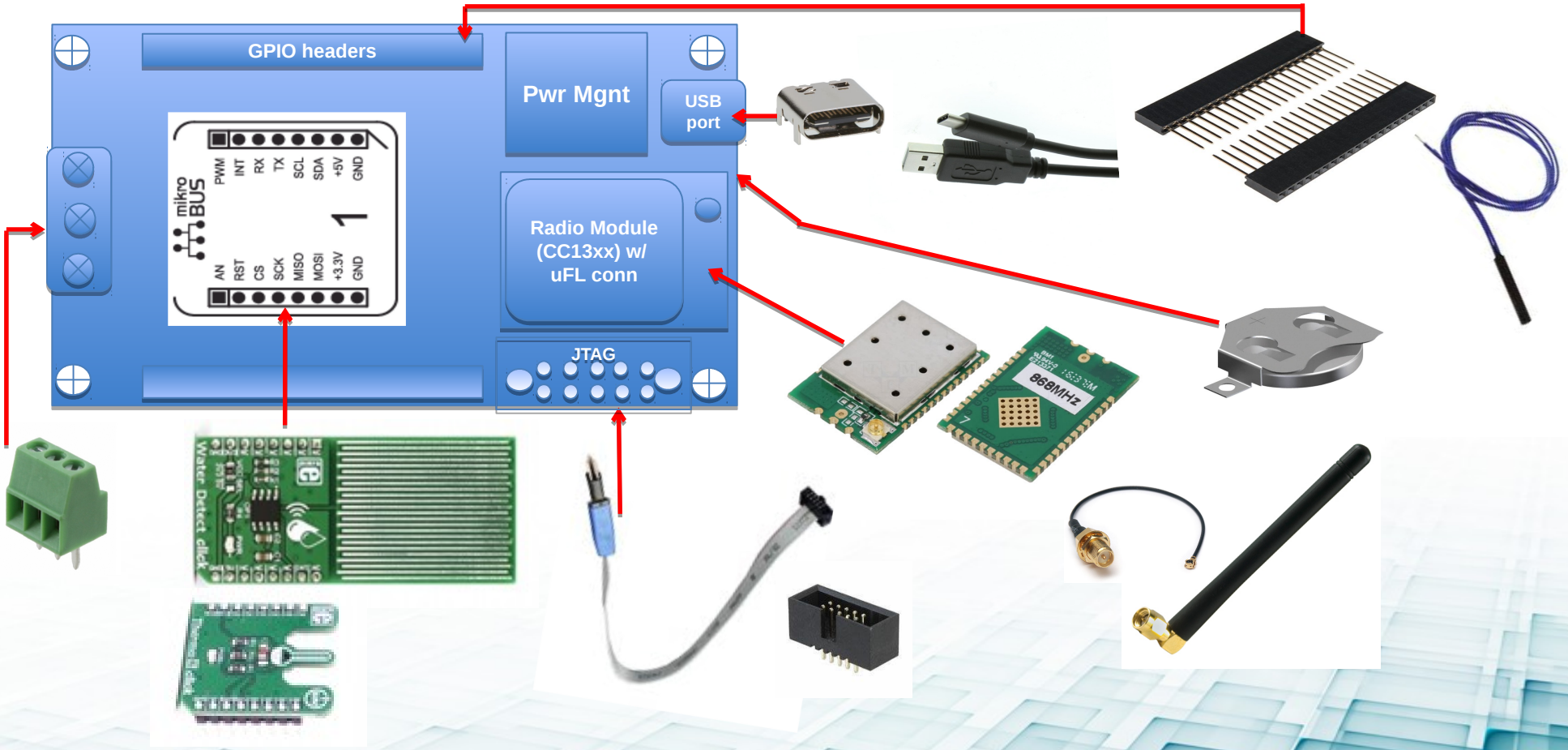
- What is probing and a probable bus?
-

Example: flood mitigation system

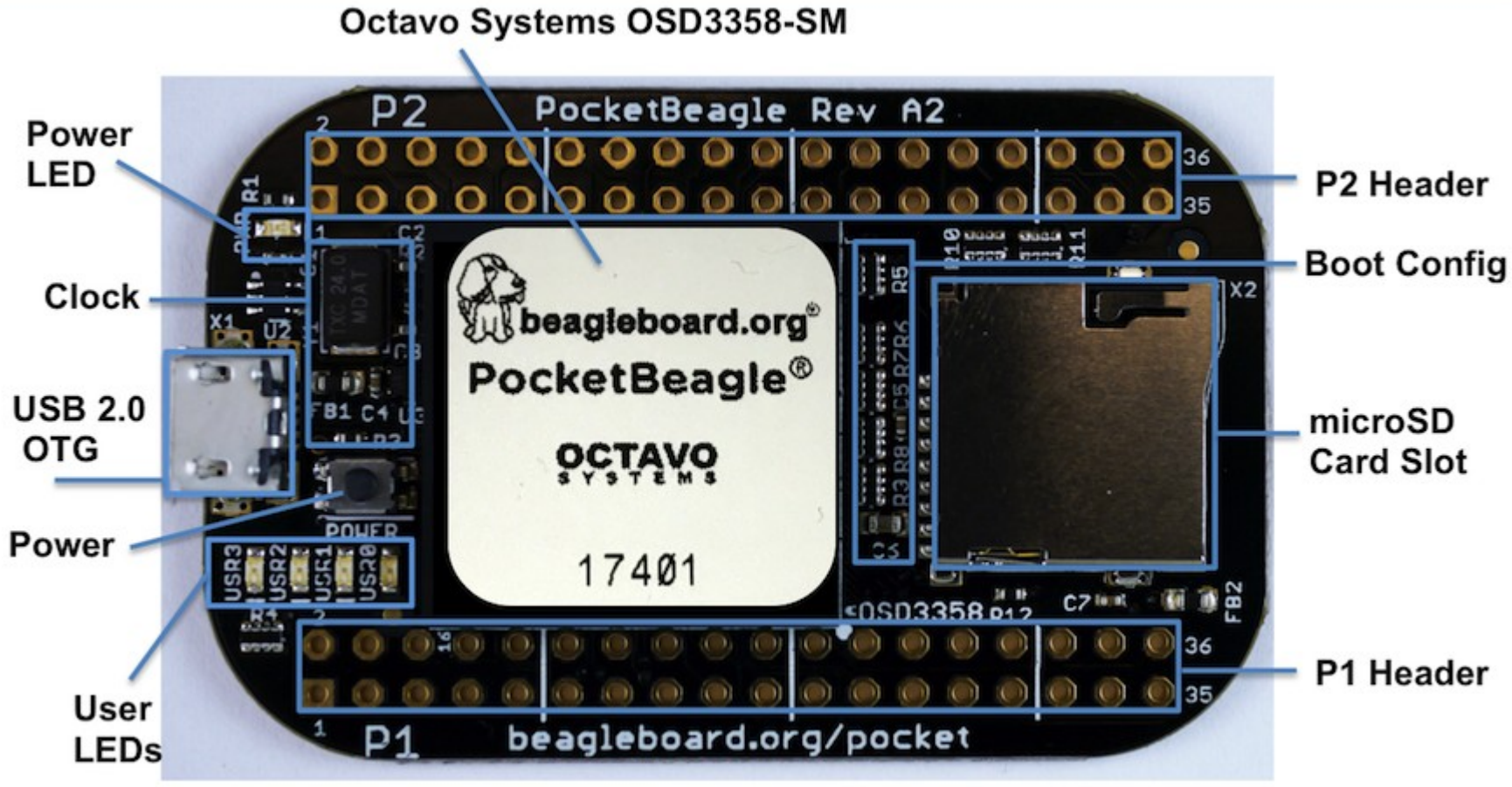
element14



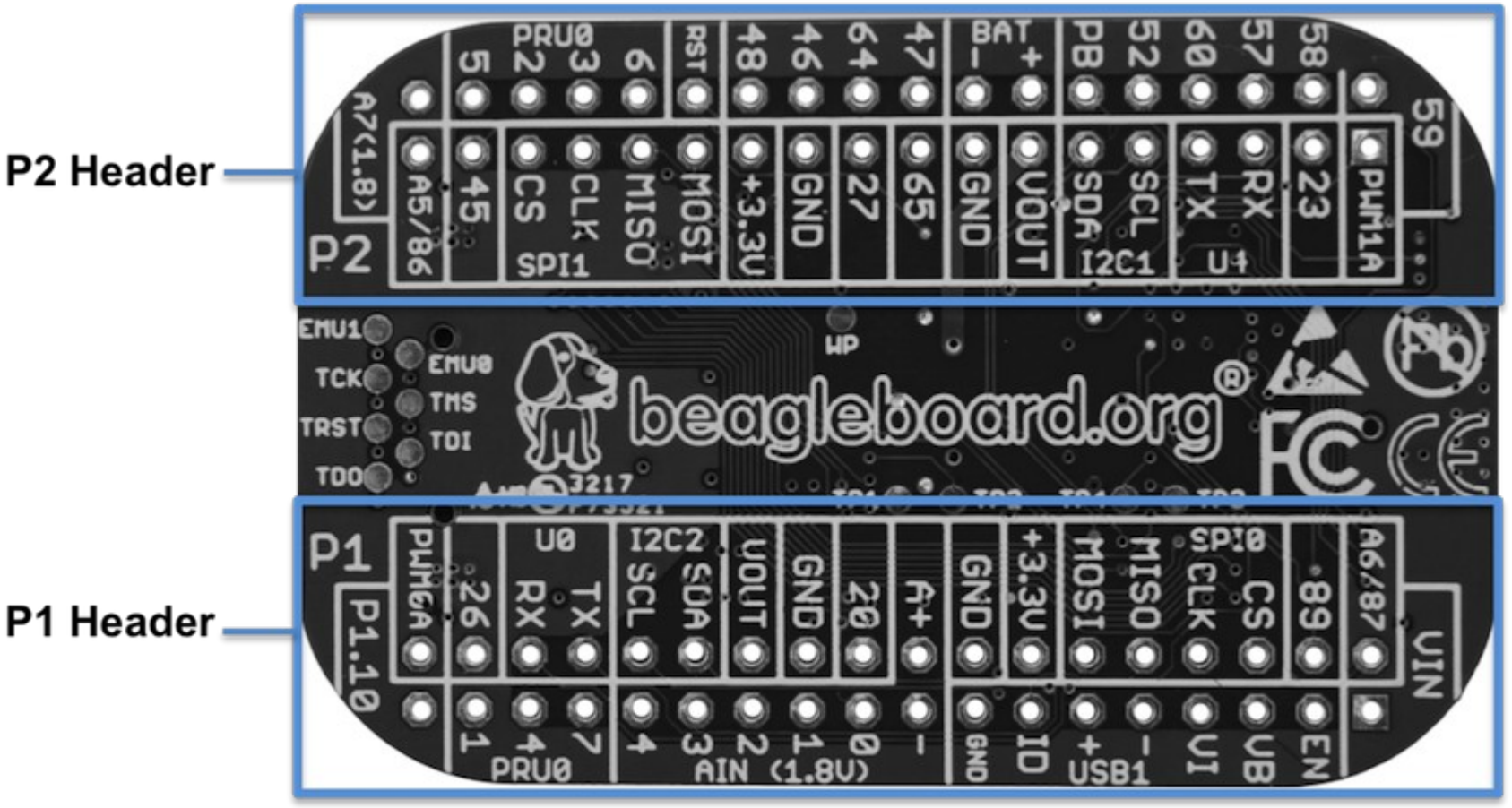
Digital Dust feature-set



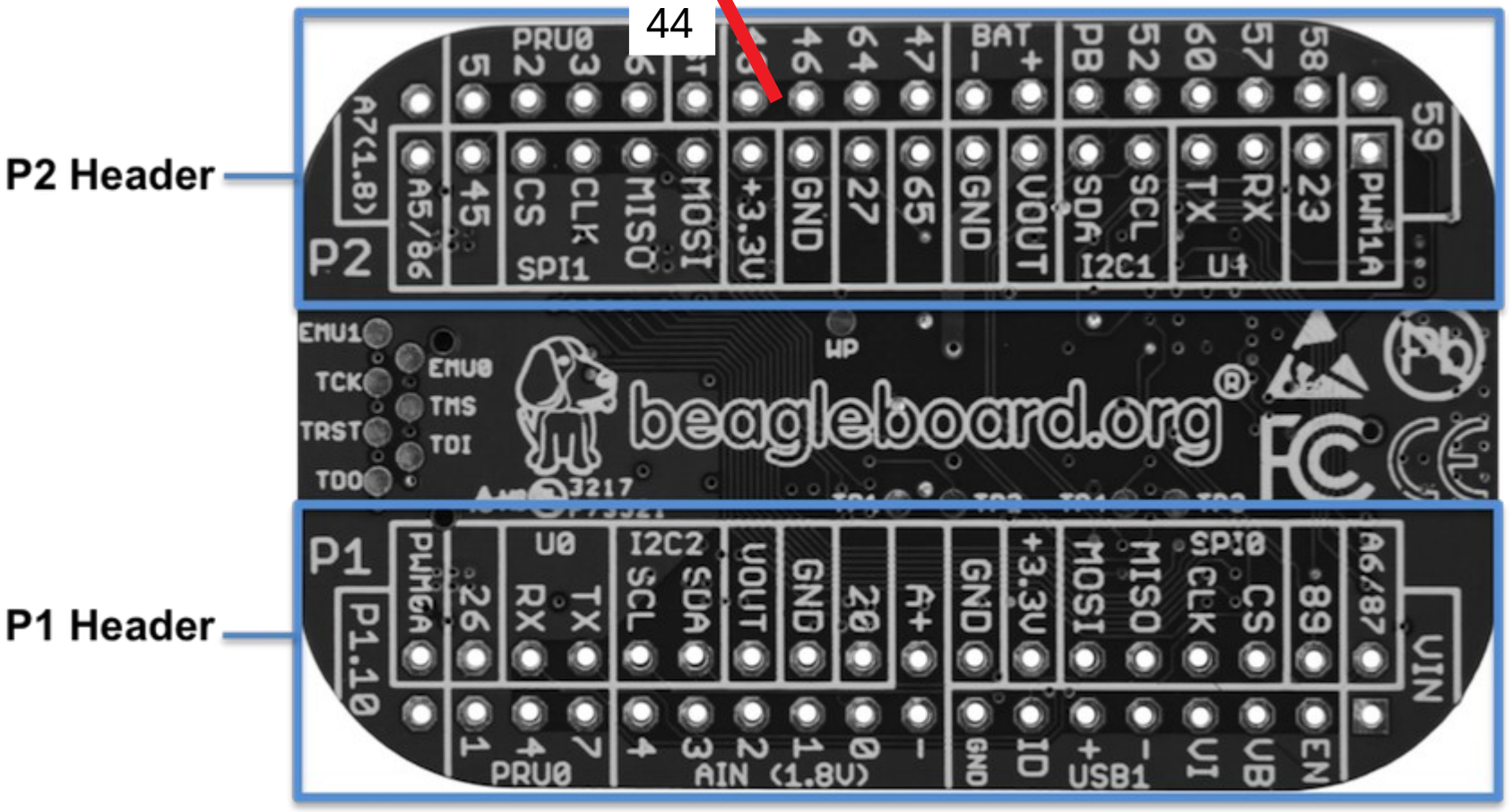
PocketBeagle top



PocketBeagle bottom



PocketBeagle bottom



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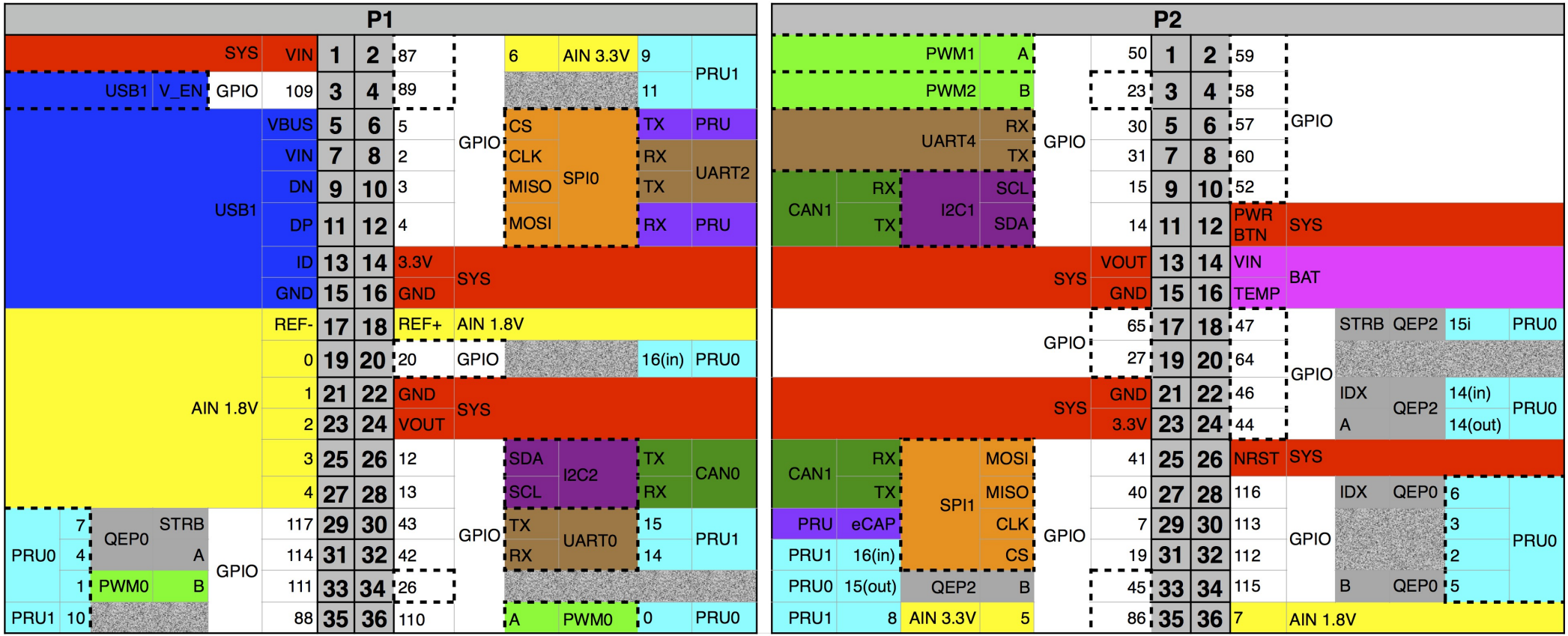
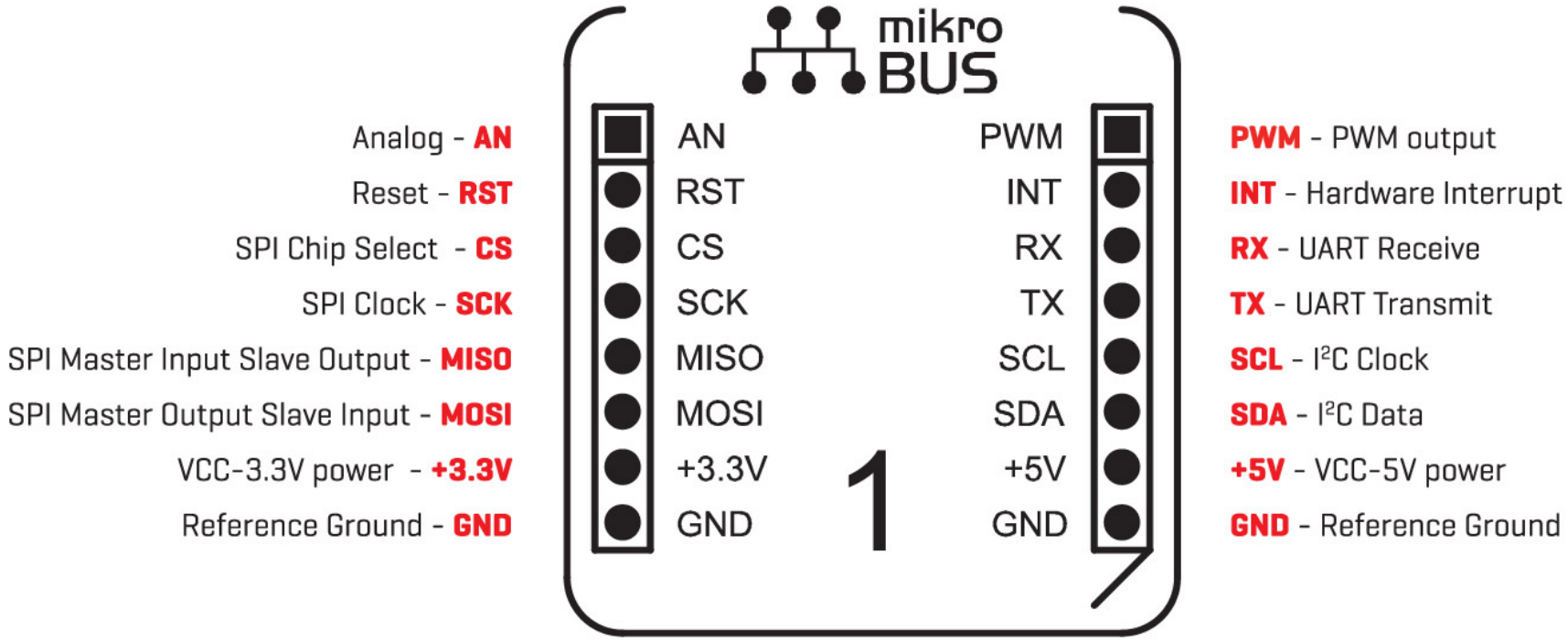
PocketBeagle expansion

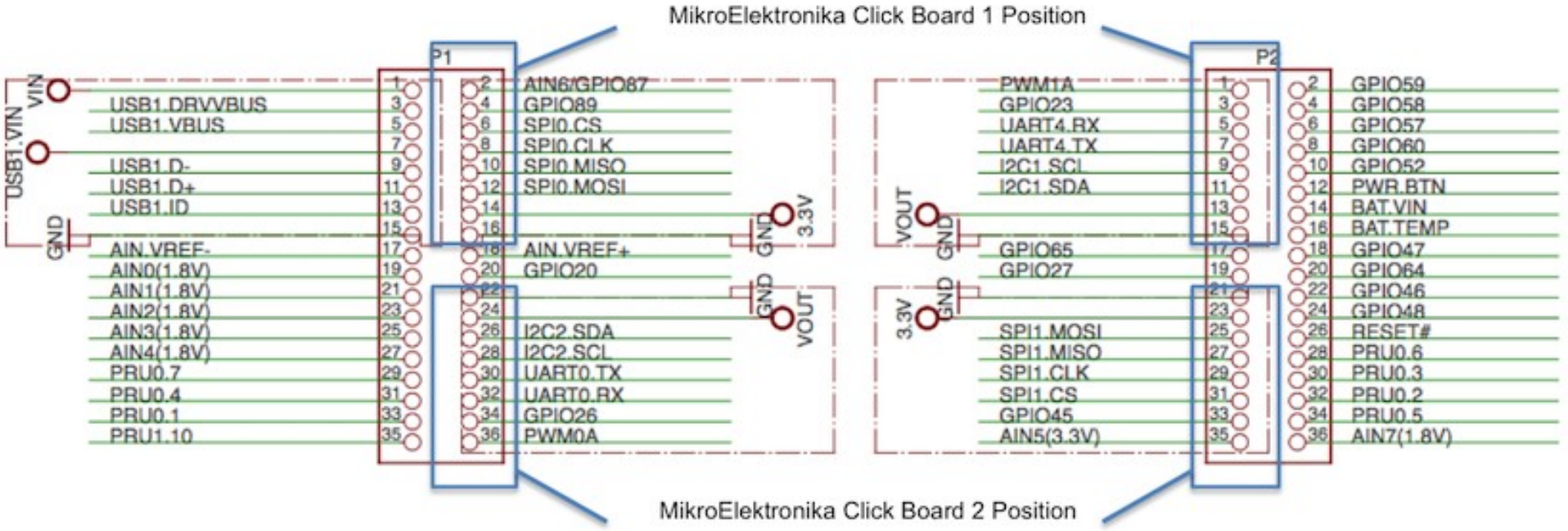
PocketBeagle Expansion Headers (Rev A2a)



mikroBus Click

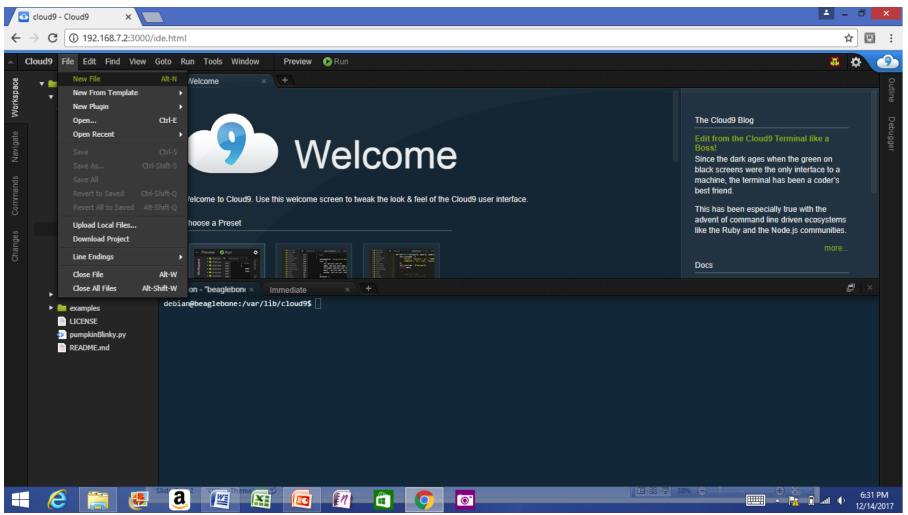


Connecting mikroBus Clicks

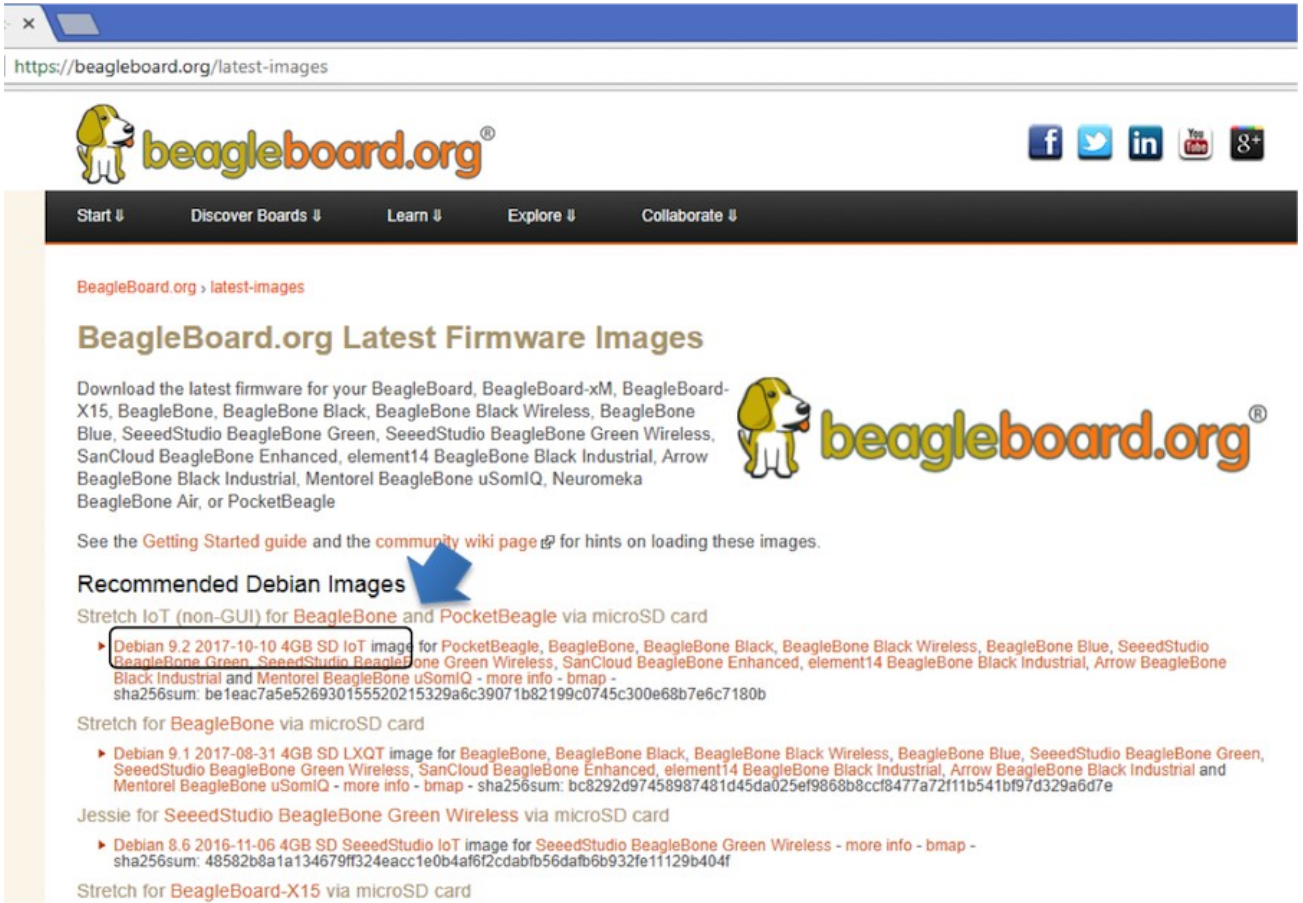


Single cable development

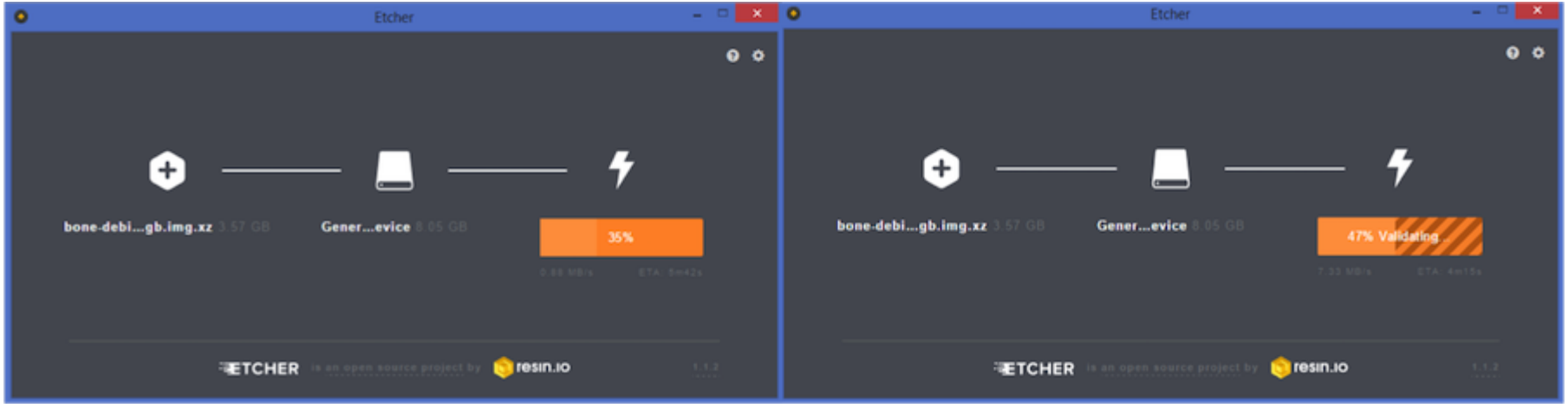
- Power, network, develop
- You can add a network and power many other ways



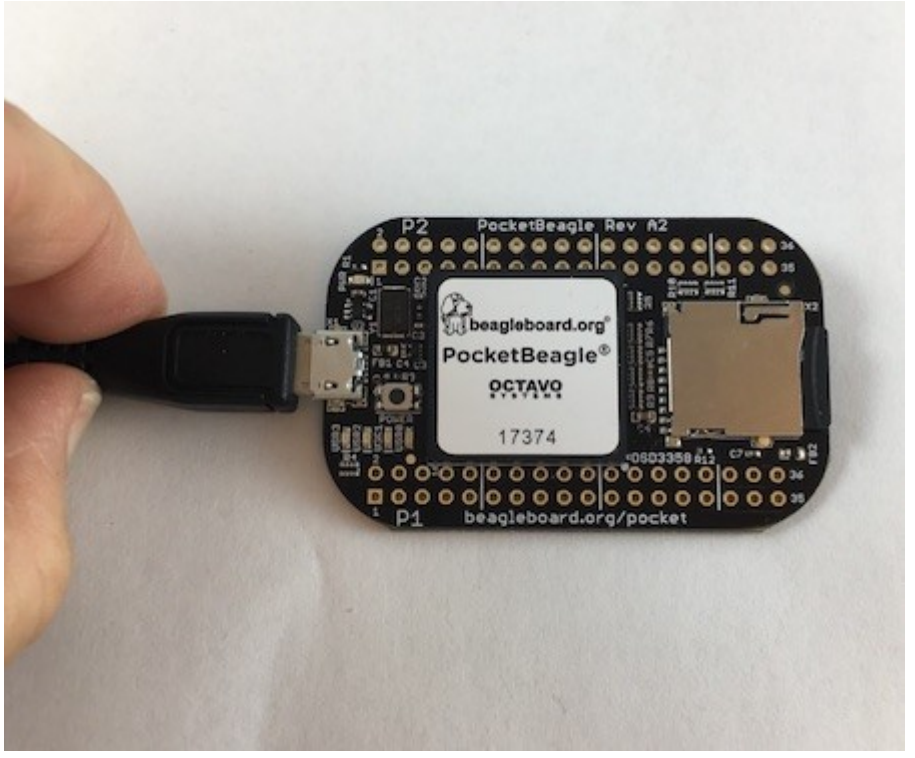
Download image



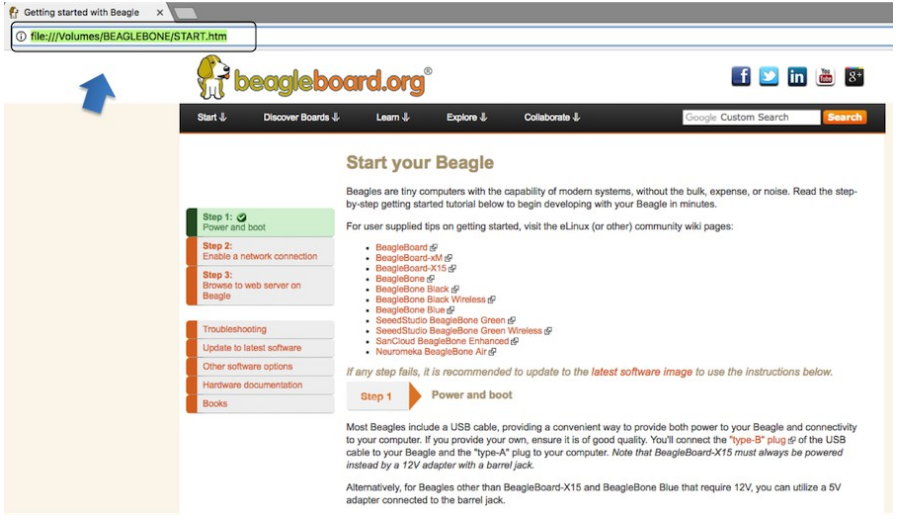
Write image to microSD with Etcher

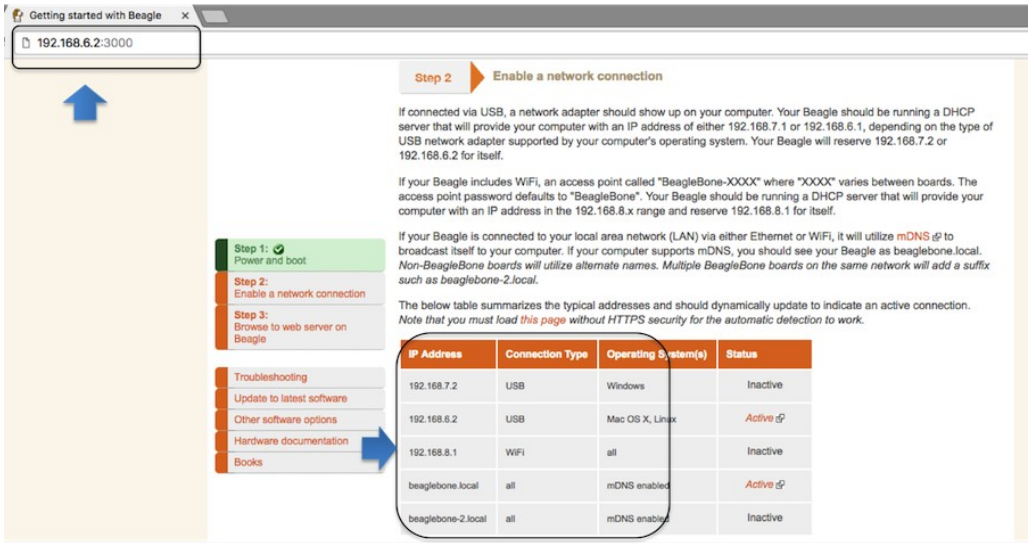


Insert microSD and boot

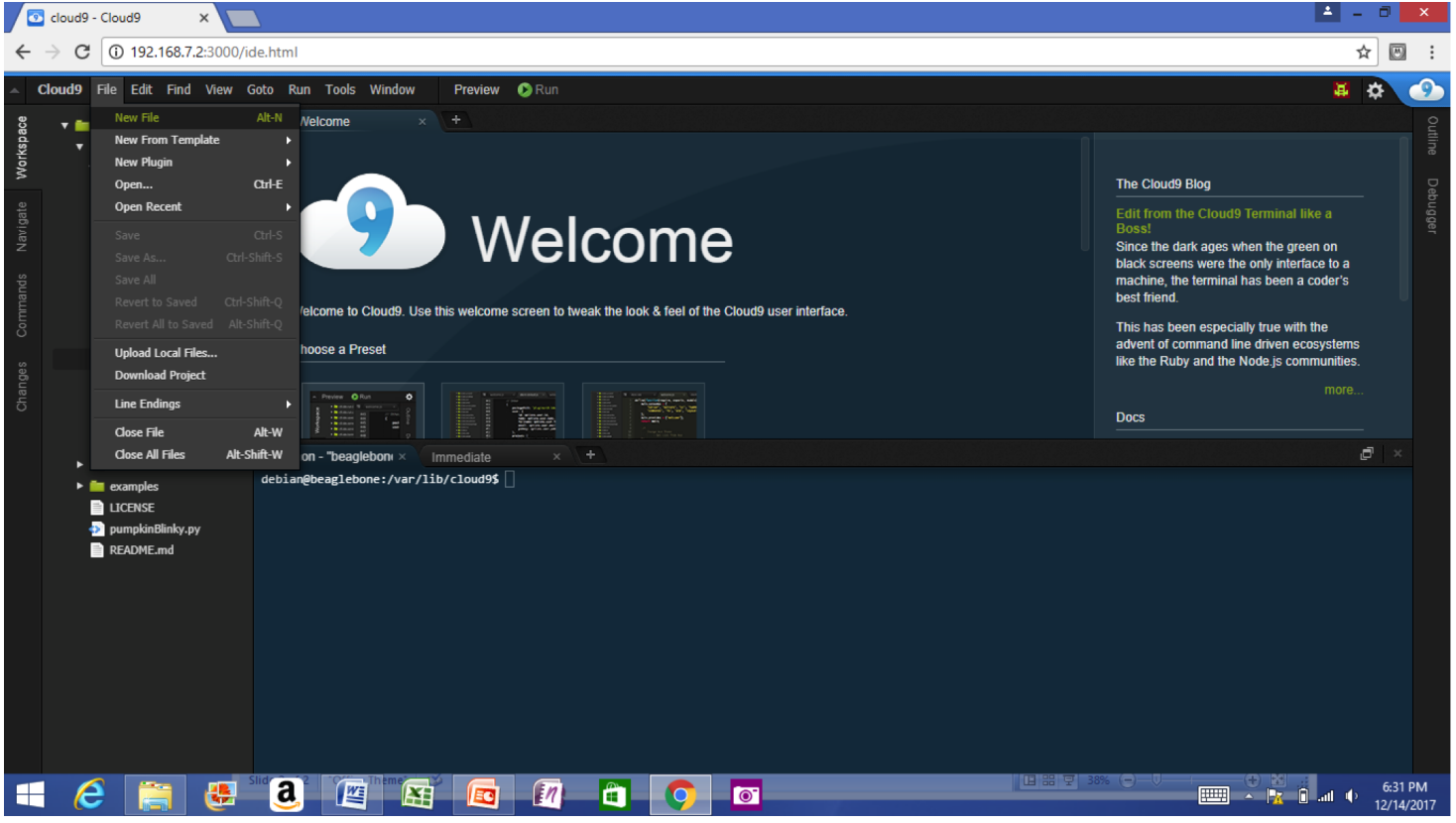


Connect to the USB network





Open the IDE


cloud9 - Cloud9
192.168.7.2:3000/ide.html
Cloud9
File
Edit
Find
View
Goto
Run
Tools
Window
Preview
Run
New File Alt-N
New From Template
New Plugin
Open... Ctrl-E
Open Recent
Save Ctrl-S
Save As... Ctrl-Shift-S
Save All
Revert to Saved Ctrl-Shift-Q
Revert All to Saved Alt-Shift-Q
Upload Local Files...
Download Project
Line Endings
Close File Alt-W
Close All Files Alt-Shift-W
Welcome
Welcome
elcome to Cloud9. Use this welcome screen to tweak the look & feel of the Cloud9 user interface.
hoose a Preset
The Cloud9 Blog
Edit from the Cloud9 Terminal like a Boss!
Since the dark ages when the green on black screens were the only interface to a machine, the terminal has been a coder's best friend.
This has been especially true with the advent of command line driven ecosystems like the Ruby and the Node.js communities.
more...
Docs
Immediate
debian@beaglebone:/var/lib/cloud9$
examples
LICENSE
pumpkinBlinky.py
README.md
Workspace
Navigate
Commands
Changes
Outline
Debugger
6:31 PM
12/14/2017

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